Student Answer Script View

MIT MPL - BTech-M Sc-MCA - II-IV and VI Semester - Midterm Examination - Mar 2025 Answer Sheet

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28.00

30.00

Ball pen refills are packed in pouches containing 25 refills in each pouch. In a shop it was found that 5 refills failed to write in pouch 1, 10 each in pouches 2 and 3, and 1 refill in pouch 4. Suppose a refill is selected at random from one of the four pouches what is the probability that it fails to write? What is the probability that it is from pouch 4?

Given:

① 4 Pouches, with 25 refills each

$\Rightarrow$ $P(P1)$ = Probability of pouch 1

$P(P2)$ = Prob of pouch 2

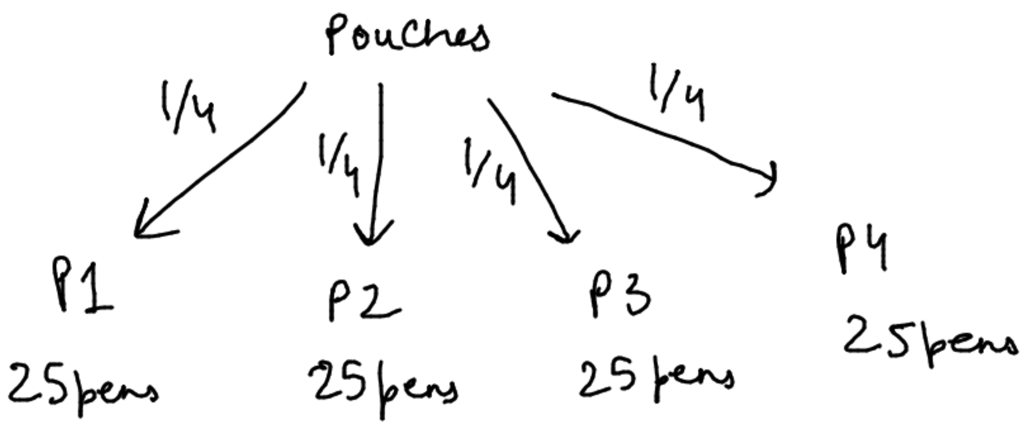
$P(P3)$ = Prob of pouch 3

$P(P4)$ = Prob of pouch 4

$\Rightarrow P(P1) = P(P2) = P(P3) = P(P4) = \frac{1}{4}$

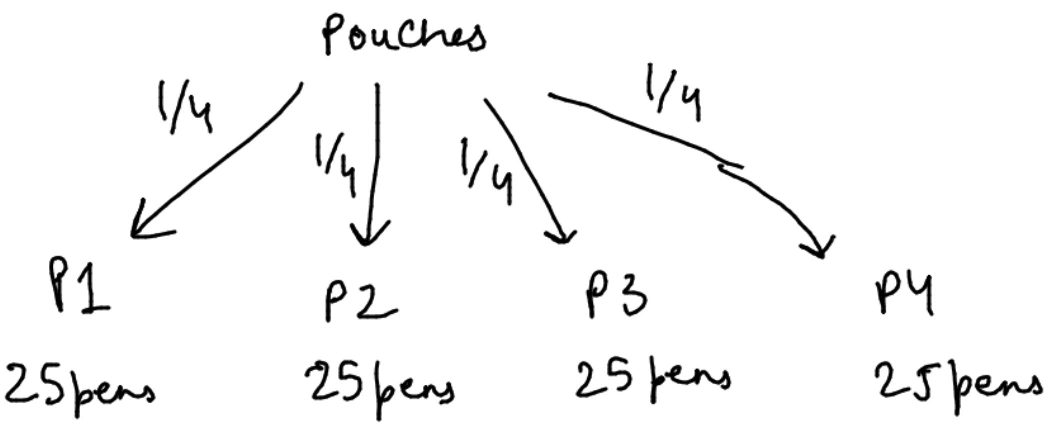
(each bag has equally likely probability)

$\Downarrow$



Let $P(F)$ denote pens that failed and $P(\bar{F})$ denote that the pen works

$\Downarrow$



$$F\left(\frac{5}{25}\right) \quad \bar{F}\left(\frac{20}{25}\right) \qquad F\left(\frac{10}{25}\right) \quad \bar{F}\left(\frac{15}{25}\right) \qquad F\left(\frac{10}{25}\right) \quad \bar{F}\left(\frac{15}{25}\right) \qquad F\left(\frac{1}{25}\right) \quad \bar{F}\left(\frac{24}{25}\right)$$

(i) <u>Probability that it fails</u> →

$$P(F) = P(F|P1)\,F(P1) + P(F|P2)\,P(P2) + P(F|P3)\,P(P3) + P(F|P4)\,P(P4)$$

$$= \frac{5}{25}\times\frac{1}{4} + \frac{10}{25}\times\frac{1}{4} + \frac{10}{25}\times\frac{1}{4} + \frac{1}{25}\times\frac{1}{4}$$

$$\Rightarrow \quad P(\text{Fails}) = \frac{13}{50}$$

(ii) <u>Probability that it failed and is from pouch 4</u> →

$$P(P4|F) = \frac{P(F|P4)\,P(P4)}{P(F)}$$

← calculated in previous question

$$= \frac{\frac{1}{25}\times\frac{1}{4}}{\frac{13}{50}} = \frac{\frac{1}{100}}{\frac{13}{50}} = \frac{1}{\cancel{100}_{\,2}} \times \frac{\cancel{50}}{13}$$

$$\Rightarrow \boxed{P(\text{Fails \& from Pouch 4}) = \frac{1}{26}}$$

Let D_n denotes the number of arrangements of $1,2,3,\ldots n$ such that the i^th element is not in i^th position. Use principle of inclusion and exclusion and show that $D_n = \frac{n!}{e}$ as $n \to \infty$.

D_n denotes no of arrangement of $1, 2, 3, ---, n$ such that i^{th} element is not in i^{th} position

Let a_1 be (1) is in correct position

$\Rightarrow N(a_1) = (n-1)!$

a_2 be (2) is in correct position

$\Rightarrow N(a_2) = (n-1)!$

- - - - - and so on till a_n

Now let $a_1 a_2$ denote both a_1 & a_2 are in correct positions

$\Rightarrow (n-2)!$

- - - - an so on till $a_{n-1} a_n$

$\therefore (\bar{a}_1 \bar{a}_2 \bar{a}_3 - - \bar{a}_n)$ implies that none of no's $1 --- n$ are in correct position

$\therefore N(\bar{a}_1 \bar{a}_2 --- \bar{a}_n) = N - N(a_1) - N(a_2) - - - -$

$--- (-1)^n N(a_1 a_2 --- a_n)$

total no. of ways is $n!$ to arrange

$$\therefore\ N(\bar{a}_1\bar{a}_2\cdots\bar{a}_n) = n! - (n-1)! - (n-1)!$$

$$- \cdots \text{(n times for all one combinations)}$$

$$+ (n-2)! + (n-2)! - -$$

$$- \cdots \text{ n times for all combinations}$$

$$+ (-1)^n \, {}^nC_n(1)$$

$$\therefore\ N(\bar{a}_1\bar{a}_2\bar{a}_3\cdots\bar{a}_n) = n! - (n-1)!\,{}^nC_1 + (n-2)!\,{}^nC_2$$

$$+ \cdots$$

$$= n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots (-1)^n\frac{1}{n!}\right)$$

$\therefore$ By principle of inclusion & exclusion

No. of derangements = $\boxed{n!\left(1 - \frac{1}{1!} + \frac{1}{2!} + \cdots (-1)^n\frac{1}{n!}\right)}$

Now as n approaches infinity

$$\lim_{n\to\infty} n!\left(1 - \frac{1}{1!} + \frac{1}{2!} + \cdots (-1)^n\frac{1}{n!}\right)$$

$$= n! \lim_{n \to \infty} \left(1 - \frac{1}{1!} + \frac{1}{2!} + \cdots (-1)^n \frac{1}{n!} \right)$$

$$= n! \times \frac{1}{e}$$

$\Rightarrow$ as $n \to \infty$

derangements $= \boxed{\dfrac{n!}{e}}$

Q.No : 3) **Score : 3.00 / 3.00**

There are two book-shelves. The first shelf has 8 Mathematics books and 4 Chemistry books. The second shelf has 10 Mathematics books and 15 Chemistry books. One of the two shelf is selected at random. Two books are then selected at random from the selected shelf. Find the probability that

(i) Both the books are Mathematics books.

(ii) One of them is Mathematics book and the other one is Chemistry book.

★ 2 Book Shelves

$\downarrow \frac{1}{2}$ $\rightarrow \frac{1}{2}$

S1	S2
8 Maths 4 Chem	10 Maths 15 Chem

★ One of the shelf is selected at random (S1 or S2)

★ Two books are then randomly selected (from selected shelf)

(i) $\boxed{\text{Both books are maths}} \rightarrow$

- Let S1 be chosen → It has 8M and 4C $\left(\begin{array}{c}\text{Total}\\ =12\end{array}\right)$

$\Rightarrow$ Total ways of choosing 2 books : ${}^{12}C_2$

- Ways of choosing 2 math books : ${}^{8}C_2$

$\therefore$ $P(\text{maths}|S1) = {}^{8}C_2 / {}^{12}C_2 = \frac{14}{33}$

- Let S2 be chosen → It has 10M & 15C $\left(\begin{array}{c}\text{Total}\\ 25\end{array}\right)$

$\Rightarrow$ Total ways of choosing 2 books : ${}^{25}C_2$

- Ways to choose 2 math books : ${}^{10}C_2$

$\therefore$ $P(\text{maths}|S2) = {}^{10}C_2 / {}^{25}C_2 = \frac{3}{20}$

$\therefore$

$$\begin{array}{ccc} & \text{Shelf} & \\ \tfrac{1}{2}\swarrow & & \searrow\tfrac{1}{2} \\ S1 & & S2 \\ \downarrow & & \downarrow \\ \text{Maths} & & \text{Maths} \\ \frac{14}{33} & & \frac{3}{20} \end{array}$$

$\Rightarrow P(\text{maths}) = \frac{1}{2} \times \frac{14}{33} + \frac{1}{2} \times \frac{3}{20} = \frac{379}{1320}$

(ii) $\boxed{\text{One is math \& other is chem}} \rightarrow$

- Shelf S1 $\rightarrow$ Total choosing ways : $^{12}C_2$

Choosing one maths & other chem : $^{8}C_1 \cdot {^{4}C_1}$

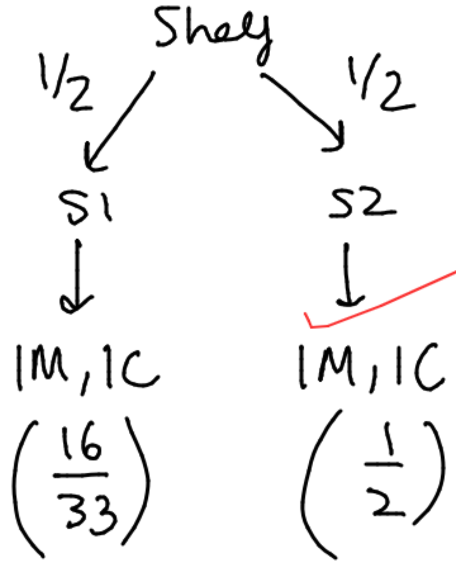
$$\therefore P(1M1C|S1) = \frac{^{8}C_1 \cdot {^{4}C_1}}{^{12}C_2} = \frac{16}{33}$$

- Shelf S2 $\rightarrow$ Total choosing ways : $^{25}C_2$

Choosing 1 Maths & 1 Chem : $^{10}C_1 \cdot {^{15}C_1}$

$$\therefore P(1M1C|S2) = \frac{^{10}C_1 \cdot {^{15}C_1}}{^{25}C_2} = \frac{1}{2}$$

$\therefore$



$$\therefore P(\text{1 Maths 1 Chem}) = \frac{1}{2} \times \frac{16}{33} + \frac{1}{2} \times \frac{1}{2}$$

$$\Rightarrow \boxed{P(1M1C) = \frac{65}{132}}$$

Q.No : 4)	Score : 3.00 / 3.00

A coin is known to come up 3 times head as often as tail. The coin is tossed 3 times. Let X denote the number of heads that appear. Write the probability distribution of X. Also find the mean and variance.

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A coin comes up with heads 3 X more like than a tails

$\Rightarrow \quad P(H)+P(T)=1 \Rightarrow 3P(T)+P(T)=1$ {given}

$4P(T)=1 \Rightarrow P(T)=\frac{1}{4}$

$\therefore P(H)=\frac{3}{4}$

- Coin is tossed 3 times.

X denotes the no. of heads $\Rightarrow$

* $\boxed{\text{Probability distribution of } X}$ $\rightarrow$

X	P(X)
0	$\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)=\frac{1}{64}$
1	$\left(\frac{3}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\times\frac{3!}{2!}=\frac{9}{64}$
2	$\left(\frac{3}{4}\right)\left(\frac{3}{4}\right)\left(\frac{1}{4}\right)\times\frac{3!}{2!}=\frac{27}{64}$

$$3 \quad \left(\frac{3}{4}\right)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{27}{64}$$

$\therefore$

X	0	1	2	3
$P(X)$	$1/64$	$9/64$	$27/64$	$27/64$

$\therefore$ Mean $(E(X)) = \sum_{0}^{3} x_i P(x_i)$

$$= 0 \times \frac{1}{64} + 1 \times \frac{9}{64} + 2 \times \frac{27}{64} + 3 \times \frac{27}{64}$$

$$\Rightarrow \boxed{E(X) = \frac{9}{4} = 2.25} \quad \text{— (1)}$$

$\star$ Variance $(V(X)) = E(X^2) - (E(X))^2$

Calculating $E(X^2) = 0^2 \times \frac{1}{64} + 1^2 \times \frac{9}{64} + 2^2 \times \frac{27}{64} + 3^2 \times \frac{27}{64}$

$$\Rightarrow E(X^2) = \frac{45}{8} \quad \text{— (2)}$$

$\therefore$ using eqn 1 & 2

$$V(X) = \frac{45}{8} - \left(\frac{9}{4}\right)^2 \rightarrow \quad V(X) = \frac{9}{16} = 0.5625$$

Hence $\rightarrow$

$$\boxed{\begin{aligned} \text{Mean, } E(X) &= 9/4 \\ \text{Variance, } V(X) &= 9/16 \end{aligned}}$$

Among all 7 digit decimal numbers, how many of them contain exactly three 9s?.

We have 7 digit numbers

⇒ _ _ _ _ _ _ _

We have to find out how many of them contain exactly 3 (9's)

⇒ We now have a total of 10 + 2 (extra 9s)
12 numbers
with 3 alike

⇒

11	12	12	12	~~12~~	12	12

(0 can't go here)

$$\Rightarrow \text{No. of ways} = \frac{\text{Totals ways}}{(\text{Alike})!} = \frac{11 \times 12^6}{3!}$$

544320

$$\Rightarrow \underset{(\text{exactly } 3\ 9\text{'s})}{\text{No. of ways}} = \frac{11 \times 12^6}{3!} = \boxed{5474304}$$

How many ways are there to distribute 25 identical balls into seven distinct boxes if the first box can have no more than 10 balls but any number can go into each of the other six boxes?.

25 identical balls into 7 distinct boxes

1 box : no more than 10 balls

& any number in the other 6

$$\Rightarrow \quad g(x) = (1+x+x^2+\cdots+x^{10})(1+x+x^2+\cdots\infty)^6$$

$$= \frac{(1-x^{11})}{(1-x)}(1-x)^{-6}$$

$$= \underbrace{(1-x^{11})(1-x)^{-7}}$$

we need to distribute 25 balls

$$\Rightarrow \quad \underset{2^{nd}\ func^n}{\overset{1^{st}\ func^n}{0+25}} \quad \text{OR} \quad \underset{2^{nd}\ func^n}{\overset{1^{st}\ func^n}{11+14}}$$

$\Rightarrow$

$$\therefore \begin{pmatrix}\text{coeff of}\\ 0 \text{ in } 1^{st}\end{pmatrix}\begin{pmatrix}\text{coeff of}\\ 25 \text{ in } 2^{nd}\end{pmatrix} + \begin{pmatrix}\text{coeff of}\\ 11 \text{ in } 1^{st}\end{pmatrix}\begin{pmatrix}\text{coeff of } 14\\ \text{in } 2^{nd}\end{pmatrix}$$

$$\Rightarrow \quad 1\underbrace{\left({}^{7+25-1}C_{25}\right)}_{{}^{n+r-1}C_r} + (-1)\left({}^{7+14-1}C_{14}\right)$$

$\Rightarrow$ ways to distribute 25 identical balls is

$$^{31}C_{25} - {}^{20}C_{14} \quad \text{ways}$$

How many $n-digit$ ternary sequences (*sequence with only* $0's$, $1's$ *and* $2's$) are there with at least one 0, at least one 1 and at least one 2?.

n digit ternary sequences with 0's, 1's & 2's

Given: Atleast one 0, Alteast one 1 and atleast on 2

$\Downarrow$

$$g(x) = \underbrace{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}_{\text{For } 0}\underbrace{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}_{\text{For } 1}\underbrace{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}_{\text{For } 2}$$

$$\Rightarrow \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)^3$$

$$= (e^x - 1)^3$$

$$= e^{3x} - 1 + 3(e^x)(-1)\left[e^x - 1\right]$$

$$= e^{3x} - 3e^{2x} + 3e^x - 1$$

$$= \left(1 + \frac{(3x)}{1!} + \frac{(3x^2)}{2!} + \cdots\right) - 3\left(1 + \frac{(2x)}{1!} + \frac{(2x)^2}{2!} + \cdots\right) + 3\left(1 + \frac{x}{1!} + \frac{x^2}{1!} + \cdots\right) - 1$$

Find coeff of $x^n/n!$ in all of these

$\Downarrow$

$$3^n - 3\cdot 2^n + 3\cdot 1^n$$

$$= \boxed{3^n - 3.2^n + 3}$$

If A and B are two independent events, show that the events

(i) $\overline{A}$ and $\overline{B}$ are independent

(ii) A and $\overline{B}$ are also independent.

Given A and B are independent events

$\Rightarrow \quad P(A \cap B) = P(A) + P(B)$

(i) $\bar{A}$ & $\bar{B}$ are also independent →

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) \quad \{\text{Demorgan's theorem}\}$$

$$= 1 - P(A \cup B) \quad \{\bar{A} = 1 - A\}$$

$$= 1 - [P(A) + P(B) - P(A \cap B)]$$

{Addition principle}

$$= 1 - P(A) - P(B) + P(A)P(B)$$

{independent events A & B}

$$= 1 - P(A) - P(B)[1 - P(A)]$$

$$= [1 - P(A)][1 - P(B)]$$

$$= \boxed{P(\bar{A})\ P(\bar{B})} \quad \{\bar{A} = 1 - A\}$$

$$\therefore \quad P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$$

Hence $\bar{A}$ & $\bar{B}$ are independent events

(ii) A & $\bar{B}$ are independent →

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$\left\{ A \cap \bar{B} = A - (A \cap B) \right\}$

$$= P(A) - P(A)\,P(B)$$

{ A & B independent events }

$$= P(A)\left[1 - P(B)\right]$$

$$= \boxed{P(A)\ P(\bar{B})}$$

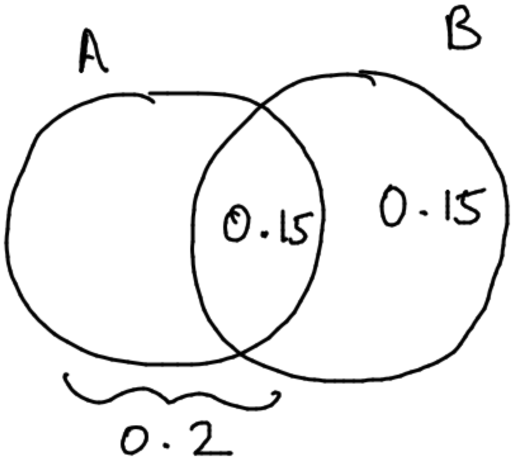
$\left\{ \bar{B} = 1 - B \right\}$

$$\therefore \quad P(A \cap \bar{B}) = P(A)\,P(\bar{B})$$

Hence A & $\bar{B}$ are independent events

An electric assembly consists of 2 subsystems A and B. From previous testing procedure the following properties are assumed to be known.
$P(A\ fails) = 0.2$, $P(B\ fails\ alone) = P(A\ and\ B\ fails) = 0.15$. Evaluate the probability that P(A fails alone).

2 subsystems, A & B



$P(A\ fails) = 0.2$

$P(B\ fails\ alone) = 0.15$

$P(A \& B\ fails) = 0.15$

$P(B - B \cap A) = 0.15$

$P(B \cap A) = 0.15$

we have to find $P(A - (A \cap B))$ {A fails alone}

$= P(A) - P(A \cap B)$

$= 0.2 - 0.15 = 0.05$

$\Rightarrow$ $\boxed{P(A\ fails\ alone) = 0.05}$

Q.No : 10)	Score : 2.00 / 2.00

A random variable X has the probability distribution function

$P\{X = k\} = \frac{c}{2^k}, k = 0,1,2,3, \ldots$

(i) Find the value of 'c'

(ii) $P\{ x \geq 5 \}$

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$$P\{X=k\} = \frac{C}{2^k}, \quad k=0,1,2,3 \ldots$$

(i)

As this a probability distribution function

Therefore,

$$\sum_{0}^{\infty} \frac{C}{2^k} = 1$$

$$\Rightarrow \frac{C}{2^0} + \frac{C}{2^1} + \frac{C}{2^2} + \ldots = 1$$

$$\Rightarrow C\left(1 + \frac{1}{2} + \frac{1}{2^2} + \ldots \infty\right) = 1$$

$\Downarrow$ This is a GP with $|r|<1$

$$\therefore \text{Sum of GP} = \frac{a}{1-r} = \frac{1}{1-1/2} = \underline{2}$$

$$\therefore C(2) = 1 \Rightarrow \boxed{C = \frac{1}{2}}$$

$$\therefore P\{X=k\} = \frac{1}{2} \times \frac{1}{2^k} = \frac{1}{2^{k+1}}, \quad k=0,1,2 \ldots$$

$$P(X \geq 5) = 1 - P(X < 5)$$

$$= 1 - \left(P(0) + P(1) + P(2) + P(3) + P(4)\right)$$

$$= 1 - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}\right)$$

$$= 1 - \frac{31}{32} \Rightarrow \boxed{P(X \geq 5) = \frac{1}{32}}$$

OR

$$\left(1/2^6 + 1/2^7 + \cdots \infty\right)$$

$$= \frac{1/64}{1-1/2} = \frac{1}{32}$$

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